

Crop Recommendation: Orange

Description:

Orange is a citrus fruit prized for its bright flavor and high vitamin C content. It thrives in subtropical climates with sandy, well-drained soils that promote healthy tree growth. Regular pruning, fertilization, and disease control measures are necessary to sustain productivity. Its steady demand in both fresh fruit and juice markets underscores its importance as a cash crop.

Growing Conditions:

- Growing Season: Subtropical (flower in spring; harvest in winter)
- Water Requirements: Moderate to high; 1000–1500 mm/year
- Optimal Temperature: 15–28°C
- Soil Preference: Sandy loam; pH 5.5–7.5

Nutritional Value:

Excellent source of vitamin C, fiber, and antioxidants

Crop Details:

Climate

- Optimal temperature: 15–28°C
- Humidity requirements: Moderate to high
- Frost sensitivity: Not frost tolerant

Soil Type

- Preferred soil: Sandy loam; pH 5.5–7.5
- Drainage requirements: Moderate
- Organic matter content: >2%

Growing Season

- Best planting time: Subtropical (flower in spring; harvest in winter)
- Duration: 3-6 months
- Crop rotation: Annual

Market Price

- Current price: \$
- Price trends: Seasonal variations
- Market demand: High

Maturity Day

- Days to maturity: nan
- Growth stages: Vegetative, reproductive, ripening
- Harvest window: 7-10 days

Water Requirement

- Moderate to high; 1000–1500 mm/year
- Irrigation frequency: Weekly
- Water depth: 5-10 cm

Nutrient Recommendation

- NPK ratio: N:80, P:40, K:60
- Micronutrients: Zinc, Iron
- Application timing: During tillering

Pest & Disease Information

- Common pests: Susceptible to citrus canker, greening
- Prevention: Crop rotation, resistant varieties